

Title: Methodology, Results, and Model Comparison Report

Project Subtitle: AI/ML Task 2 — California Housing Price Prediction System

Section 1: Executive Summary

This report presents the implementation and empirical evaluation of a machine learning system designed to predict median house values using the California Housing dataset. The primary goal is to establish a rigorous, feature-scaled engineering pipeline to benchmark three distinct regression algorithms: Linear Regression, Ridge Regression, and a Decision Tree Regressor. Model performance is validated using Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2 Score).

Section 2: Technical Methodology & Pipeline Workflow

To ensure industry-standard machine learning practices, a structured five-stage engineering pipeline was built:

1. **Dataset Ingestion:** The dataset was loaded via `fetch_california_housing(as_frame=True)`, combining 8 numeric attributes (including median income, housing age, and average rooms) with the continuous target variable (`HousePrice`).
2. **Feature Scaling (Critical Step):** Because features have completely different numerical ranges (e.g., population counts vs. fractional room averages), standard normalization was executed using `StandardScaler`. This transforms the data to have a mean of 0 and a standard deviation of 1, preventing high-magnitude columns from artificially dominating the model weights.
3. **Data Splitting:** The data was partitioned into an 80% training set and a 20% validation testing set using a fixed random state (`random_state=42`) to guarantee absolute replication of benchmarks.

Section 3: Predictive Model Architecture Rationale

Three model paradigms were tested to expose trade-offs across bias, variance, and non-linear adaptability:

- **Linear Regression:** Serves as the baseline model, assuming strict linear dependencies between housing features and prices.
- **Ridge Regression (L₂ Regularization):** Adds an alpha penalty ($\alpha=1.0$) to the squared weights of the linear equation to counter potential multi-collinearity and prevent overfitting.
- **Decision Tree Regressor:** Introduced as a non-parametric alternative restricted to a maximum depth (`max_depth=5`) to map non-linear spatial and structural patterns without over-memorizing training data.

Section 4: Empirical Results & Performance Comparison

After training on the 80% data slice, the candidate models produced the following exact metrics on the unseen 20% test set:

Model Architecture	RMSE (Lower is Better)	R ² Score (Higher is Better)	Status
Linear Regression	0.7456	0.5758	Baseline. Model
Ridge Regression	0.7456	0.5758	Regularized Baseline
Decision Tree (depth=5)	0.7362	0.5864	Best performing model

Section 5: Final Conclusion & Insights

1. **The Winning Architecture:** The **Decision Tree Regressor** achieved the highest accuracy ($R^2 = 0.5864$) and the lowest average error $\{RMSE\} = 0.7362$.
2. **Why it Succeeded:** Housing prices are heavily influenced by non-linear factors (such as geographical locations mapped via Latitude/Longitude combined with localized income wealth). Tree-based branching naturally uncovers these non-linear boundaries, whereas linear models are restricted to flat mathematical hyperplanes.
3. **Linear vs. Ridge Behavior:** Both linear models returned identical scores, indicating that severe multi-collinearity was not present in the data split.